

Scalability & Future Plan

COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

SMARTBRIDGE

SKILL WALLET PRODUCT

DATE 29 September 2026	TEAM ID UNM123APEXLAB	PROJECT NAME ComicCraft	MAX MARKS 1 Mark
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Executive Overview: This document outlines how the **ComicCraft** platform can be scaled from a local FastAPI setup into an enterprise-grade multi-modal generative application. It defines strategies for managing heavy compute demands (Gemini models and Stable Diffusion image workflows), state persistence, and continuous feature expansion.

1. CURRENT SYSTEM LIMITATIONS

The current implementation of ComicCraft uses a synchronous processing workflow designed for local evaluation. The following bottlenecks have been identified for high-concurrency production workloads:

S.No	Limitation	Impact	Priority
1	Synchronous Sequential Execution	Generating 5 panels sequentially (Gemini Flash outline → Gemini Pro narration → Stable Diffusion image rendering per panel) creates long user HTTP wait times and blocks server threads.	High
2	Stateless / No Data Persistence	The application operates without user accounts or database integration. Users cannot save, revisit, or manage previously generated comics after closing the preview session.	Medium
3	Local Ephemeral File Storage	Panel images and compiled PDF exports are written directly to local disk folders (/static/panels, /static/exports). This leads to rapid storage exhaustion and prevents horizontal scaling across multiple instances.	High

2. SCALABILITY PLAN

To transition from local execution to high-availability web scale, the following architectural upgrades are planned:

S.No	Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single FastAPI instance running via Uvicorn; handles synchronous single-user HTTP requests locally.	Containerize with Docker and orchestrate on Kubernetes / AWS Elastic Beanstalk with auto-scaling policies behind an Application Load Balancer.
2	Data Storage	Local filesystem storage for generated images, static assets, and generated PDF documents.	Migrate asset storage to AWS S3 / Google Cloud Storage CDN, and implement PostgreSQL/MongoDB for persistent storing of user profiles and comic metadata.

2. SCALABILITY PLAN (CONTINUED)

S.No	Aspect	Current State	Proposed Upgrade / Solution
3	Performance	In-process Diffusers pipeline execution and synchronous sequential loops for AI model calls.	Implement asynchronous worker queues (Celery + Redis) to parallelize panel generation and offload GPU tasks to specialized inference worker pools.
4	Security	Direct API key configuration in local .env file with unauthenticated route endpoints.	Implement JWT/OAuth2 user authentication, API rate-limiting via FastAPI-Limiter, and centralized secrets management using AWS Secrets Manager.

3. FUTURE ROADMAP

The ongoing development plan for ComicCraft focuses on enhancing visual consistency, introducing social sharing, and extending storytelling capabilities over multi-phase milestones:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Accounts & Personal Library	Month 1 – 2	Enables user authentication, allowing creators to view, organize, edit, and re-download previously generated comics from a personal library.
Phase 3	Async Queues & Character Consistency	Month 3 – 4	Eliminates HTTP timeout issues during creation and introduces fine-tuned LoRA / ControlNet models for continuous character consistency across comic panels.
Phase 4	Multi-Page Arcs & Community Feed	Month 5 – 6	Allows expanding single-strip comics into full multi-page comic books alongside a public community showcase with sharing, liking, and commenting options.

Summary: By implementing background task queuing, cloud storage, persistent database models, and character fine-tuning, ComicCraft transitions seamlessly from a functional prototype to a scalable generative AI platform for creative storytelling.